ACS-3922 Fall 2026

Assignment 3

Due Date: Monday, November 16 at 11:59 pm

Part A – Complete Tutorial

Complete Tutorials 4 such that you have a complete Top Down Shooter from which to build from

Part B – Build a Game

Using what you built in Tutorial 4 as the basis, create **one** of the following games:

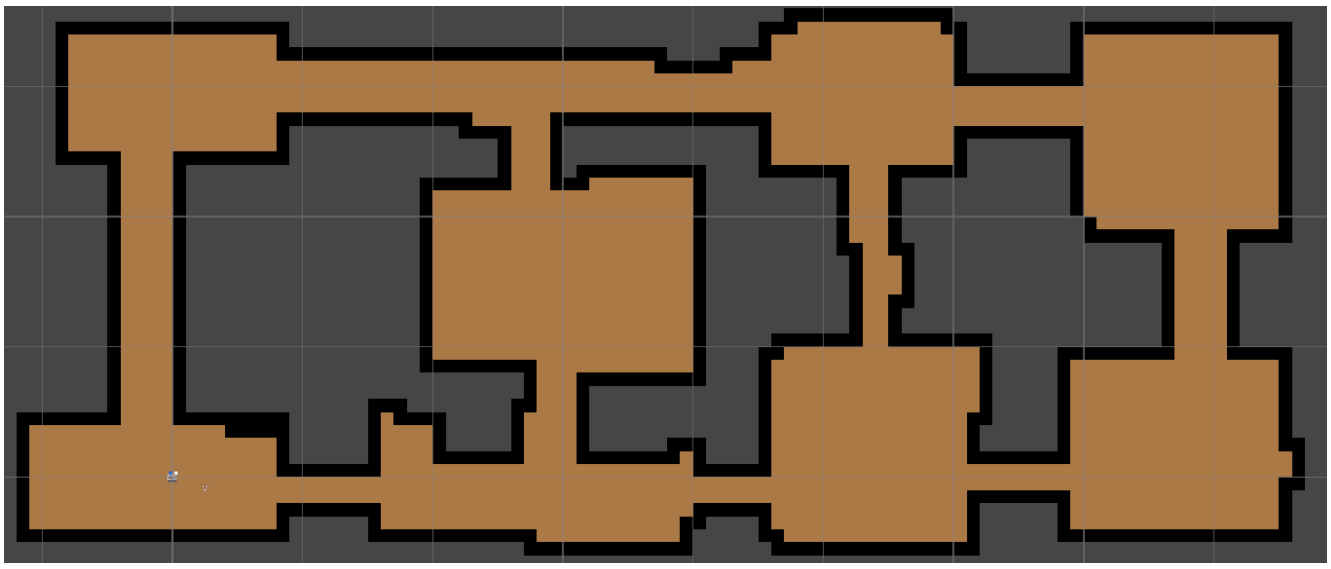
1) Dungeon Shooter

Design:

For this design, you will create a basic dungeon crawler that uses our top down shooter framework.

Requirements:

- Use a tilemap to generate rooms and corridors for the player to explore.
- You must implement a camera controller to follow the player (see Tutorial 3)
- Your dungeon must have at least 2 floors.
- Walls should be made using a tilemap
- Each floor of your dungeon must occupy a space at least 57 tiles wide x 30 tiles high.
    - Your map doesn't need to fill every square inch of this area (i.e. you can have larger rooms connected by smaller corridors), but the foot print of the entire map should at least cover the area.



An example of an acceptable map layout.

- You must implement exits that takes you from the first floor to the second floor, and from the second floor to the game end screen.
- You must implement a health system for the player
    - Player has a maximum amount of health
    - Health is subtracted from the player when they take damage
    - When the player's health drops below 1, they are killed and the game is over.
- Display the player's current health in HUD.
- You must implement a hazard that damages the player when they walk over it
    - The hazard should damage the player when they first step onto the hazard
    - The hazard should then damage the player after a set time interval if they continue to stand on or walk over the hazard
- You must implement a gold collection system for the player
- Display the player's current gold amount in the HUD.
- You must implement a gold pickup that adds to the player's gold total.
- You must implement a health pickup that replinishes the player's health.
- You must have crates that have a chance of dropping health or gold items when they are destroyed.
- You must implement at least 1 additional enemy type that demonstrates some different or additional behaviour than the enemy provided in the tutorial.
- You must have a game over screen for when the player dies.
- You must implement a game over screen for when the player escapes the dungeon and wins which displays the total gold collected.

2) Box Pushing Shooter

Design:

Sokobans are block pushing games where you push blocks around to get them into a specific alignment and solve spatial puzzles. For this design, you're going to take the top down shooter framework and turn it into a Sokoban where you need to shoot the blocks in order to get them to move.

Requirements:

- Use tilemap to generate walls for your levels
- You game must have at least 4 different levels to solve
- The challenge of the puzzles in your levels must increase with each level
- Each stage must be at least 38 tiles wide and 10 tiles high, or 19 tiles wide and 20 tiles high
    - You can mix and match wider and taller stages over your 4 required stages
- You must implement a camera controller to follow the player (see Tutorial 3)
- Levels are made up of blocks that need to be pushed into specific places
- You must visually indicate where the blocks need to go to solve the level
    - for example: have an icon on the top of the box that matches an icon on the floor where the box needs to be pushed to.
- Blocks are pushed by shooting them.
- When you shoot a block, it moves a predetermined distance.
- Block movement must be restricted to the 4 cardinal directions: North, South, East and West.
    - i.e. If I shoot a block from at an angle, and the laser hits the South side, the block will move

straight North. It doesn't need to move along the angle of the shot.

- You must add feedback for when a block is shot.
- You must have Enemies that patrol areas (you can use the patrol logic from the tutorial)
    - If the player contacts an enemy, the level is reset
    - Enemies cannot be shot
- Add a HUD and track the number of moves made by the player (each push of a block is a move).
- You must add a "Reset Stage" button to your HUD. When pressed, it resets the current stage to it's initial state, and resets the player's move count for that stage.
- At the end of every stage and the end of the game, display a screen showing the player the number of moves it took to beat that stage, along with the total number of moves the player has used in all stages.

3) Stealth Shooter

Design:

For this design, you'll turn our top down shooter into a top down stealth game. In stealth games, the player must use stealth and reconition of enemy movement patterns to avoid combat and escape.

Requirements:

- Your game must have 3 different stages.
- Walls should be made using a tilemap.
- Each stage must be at least 38 tiles wide and 10 tiles high, or 19 tiles wide and 20 tiles high
    - You can mix and match wider and taller stages over your 3 stages
- You must implement a camera controller to follow the player (see Tutorial 3)
- The tutorial enemy must be modified so that it only detects the player when they are within a cone of detection in front of the enemy and the enemy has an uninterrupted line of sight.
    - HINT: use raycasting to detect if the enemy has uninterrupted line of sight when the player is within the detection area.
- You must add visualization for the enemy detection cone (this can be as simple as a semi-transparent sprite in the shape of the detection area attached to the enemy)
- When the player is detected by the enemy, they loose a life and must restart the level.
- You must implement a lives system where the player has a minimum of 3 lives.
- You must add a HUD with the current number of player lives and the current stage.
- You must modify the player shooting so that the player can only fire a limited number of times.
- You must indicate how many shots the player has remaining on the HUD.
- You must modify the player shooting so that instead of damaging enemies, shots from the player only renders them unconcious for a set duration of time.
    - While unconcious, guards cannot move or detect the player
- You must add a exits that will take you from one level to the next, and finally to the end game screen.
- You must have a game over screen with a winning message when the player wins the game (escapes all the stages), and a game over sceen with a loosing message when the player looses the game (all lives lost).

Part C – Game Feel

Create a new markdown file called A3PartC.md.

1) Be sure to cite any sources that you may have used for your game.
2) Identify which of the 3 games designs you have decided to create.
3) Give your game a title and a story: this can be very simple. A couple of senteces that establish: the world of the game, who the main character is, and the goal of the game.
4) Write a paragraph on what Game Feel you wanted to achieve with this game. Explain how the game feel you were designing for relates to your narrative or theme.
5) Write a paragraph on what you did in your game to achieve the Game Feel you were aiming for. Identify what parts of input, response, context, polish, and metaphor you manipulated, and how you manipulated them in order to try and achieve your desired game feel.